

Outlier Detection & Handling

A Beginner's Guide to Finding and Managing Data Anomalies

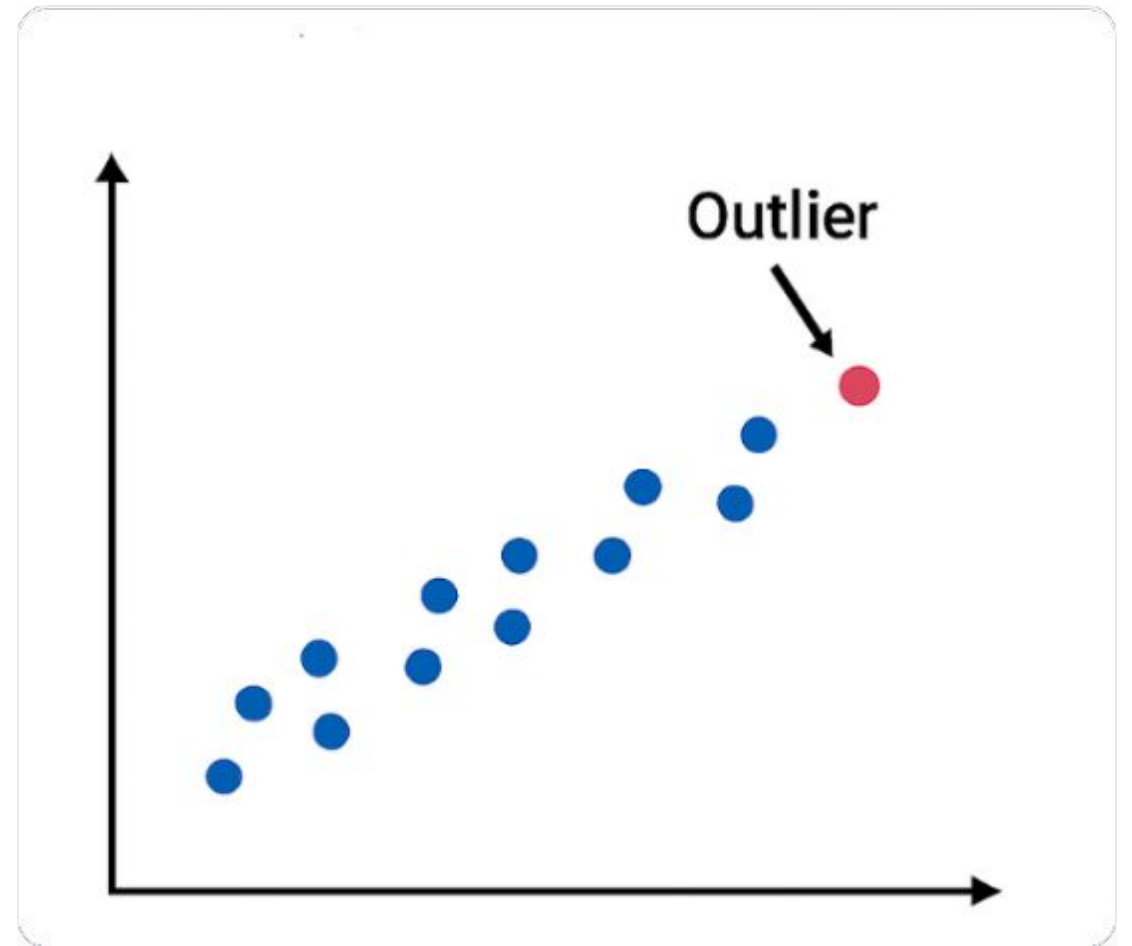


| What is an Outlier?

An **outlier** is a data point that differs significantly from other observations in your dataset.

Think of it as the "odd one out." It could be a billionaire moving into a middle-class neighborhood, or a sudden spike in website traffic due to a viral post.

Outliers can be caused by genuine variability in the real world, or they can simply be data entry errors.



Why Do Outliers Matter?



Skewed Averages

A single extreme value can drastically pull the "Mean" (average) up or down, giving a false impression of what is considered normal.



Model Disruption

Machine learning models try to find general patterns. Outliers confuse algorithms, causing them to make less accurate predictions.



Hidden Insights

Sometimes, the outlier IS the insight! In fields like fraud detection or medical diagnosis, the anomaly is exactly what you want to find.



1. Detection Methods

How do we mathematically identify the anomalies?

| The Z-Score Method

The Concept

The Z-score measures exactly how many **Standard Deviations** a data point is away from the mean.

If your data follows a normal "Bell Curve" distribution, nearly all normal data sits close to 0. Anything too far out is flagged as an outlier.

The Formula

$$Z = \frac{x - \mu}{\sigma}$$

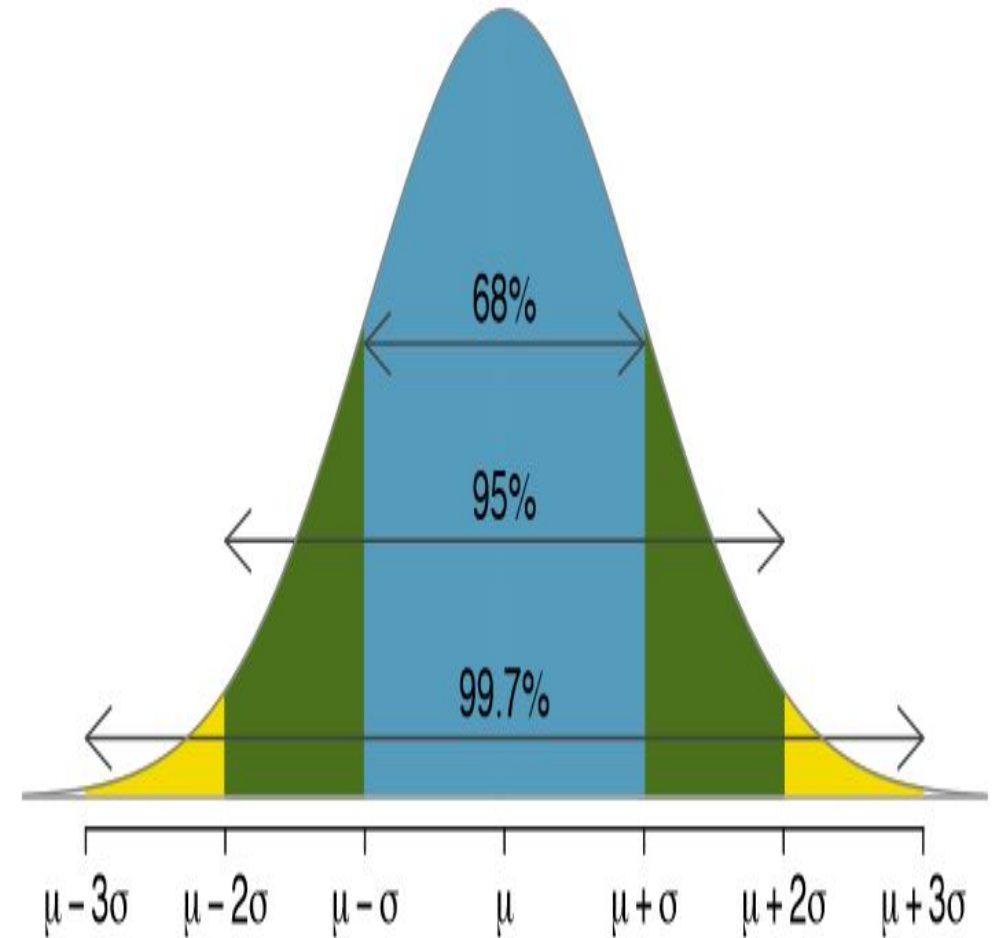
x = Data Point | μ = Mean | σ = Standard Deviation

Visualizing Z-Scores

In the Z-score method, we typically use a threshold of **+3 or -3**.

- ✓ If **$Z > 3$** , the value is abnormally high.
- ✓ If **$Z < -3$** , the value is abnormally low.

Because 99.7% of data in a normal distribution falls within 3 standard deviations, anything beyond these "tails" is highly unusual.

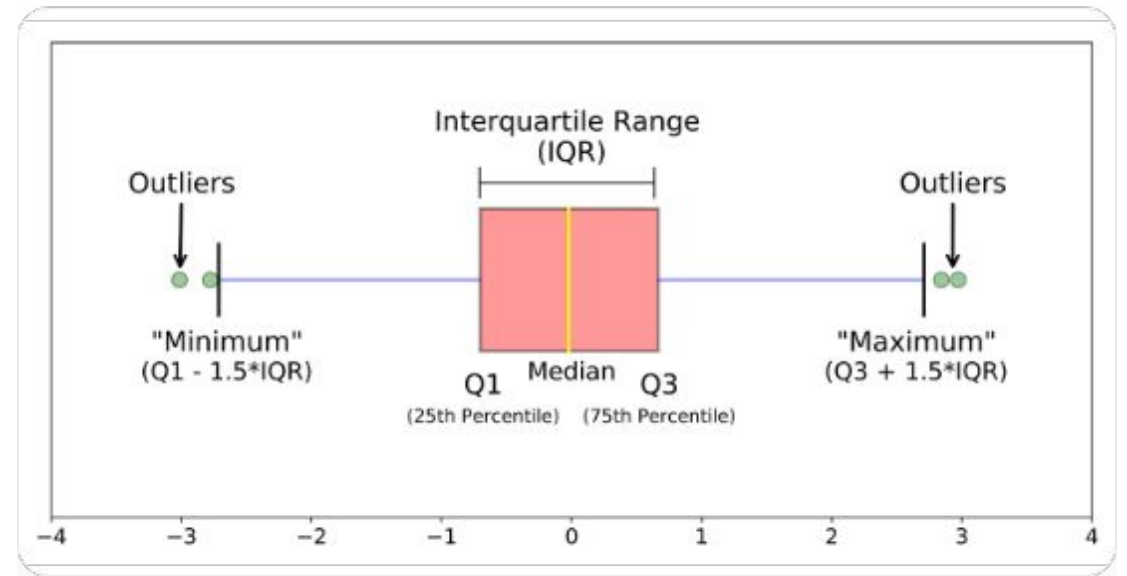


The IQR Method

The **Interquartile Range (IQR)** focuses on the middle 50% of your data.

Unlike the Z-score, the IQR method does NOT require your data to follow a perfect bell curve. It is based on medians and percentiles, making it highly robust against extreme values.

We visualize this method using a **Box Plot**.



| The 1.5x Multiplier

1.5

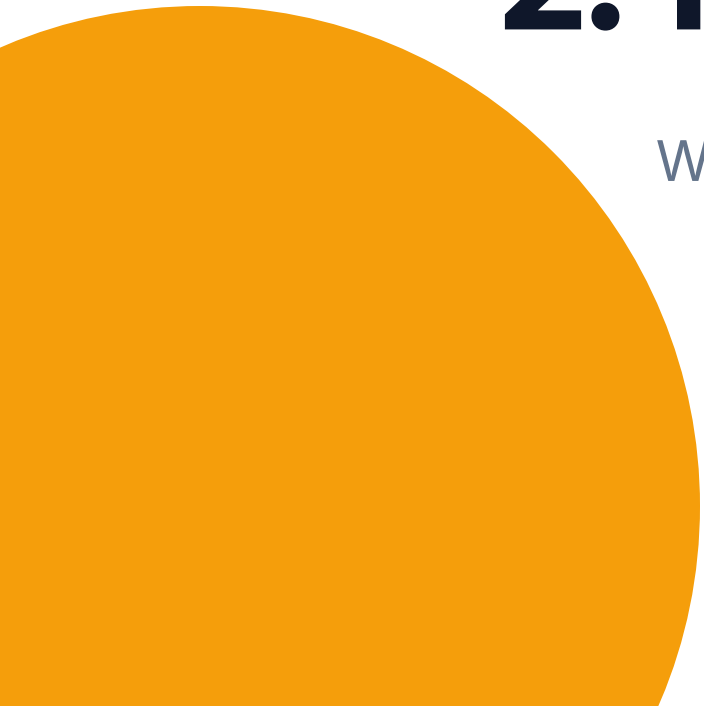
The Magic Constant

Setting the "Fences"

To detect outliers using IQR, we build invisible boundaries (fences) around our data:

- ✓ **Lower Bound:** $Q1 - (1.5 \times IQR)$
- ✓ **Upper Bound:** $Q3 + (1.5 \times IQR)$

Any data point that falls below the lower bound or above the upper bound is officially classified as an outlier.



2. Handling Methods

We found them. Now what do we do with them?

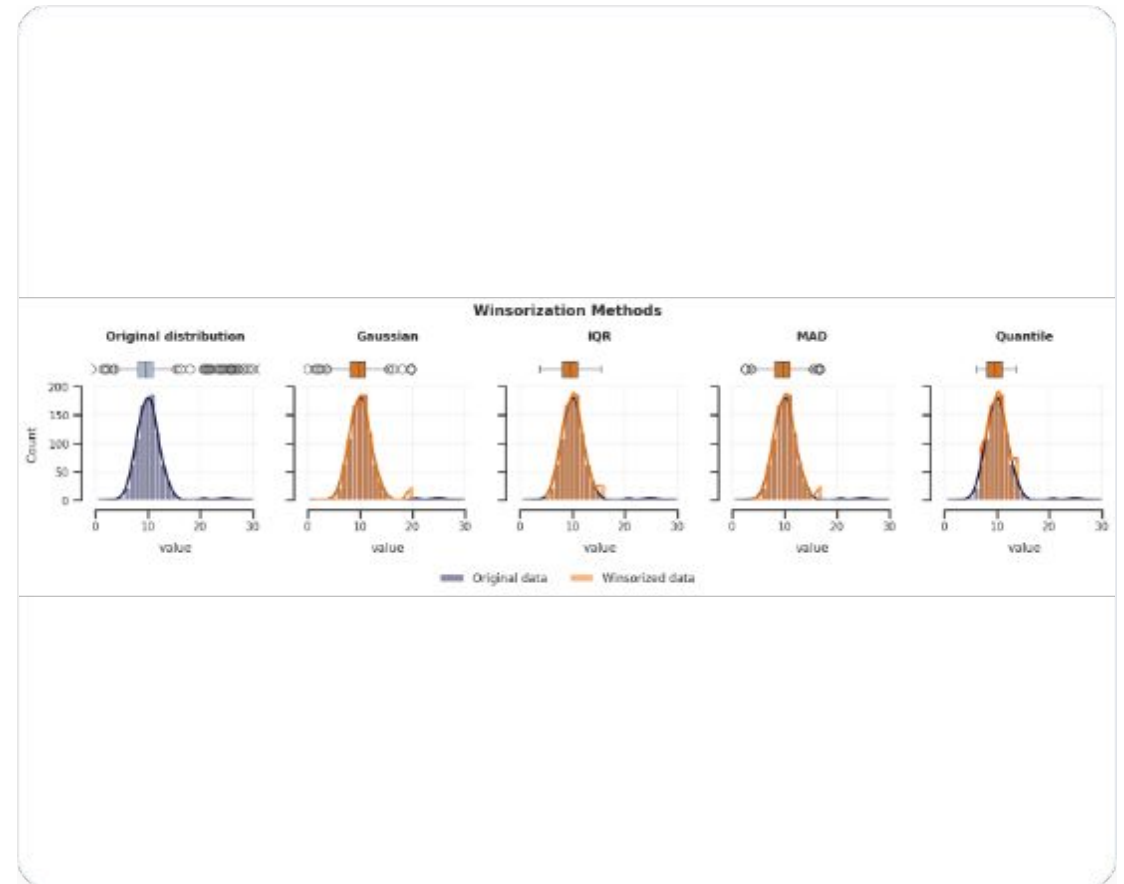
The Winsorization Method

Deleting outliers means losing data. **Winsorization** offers an alternative: we "cap" the extreme values.

For example, if we cap data at the 1st and 99th percentiles:

- ✓ Any value above the 99th percentile is replaced by the exact 99th percentile value.
- ✓ Any value below the 1st percentile is replaced by the 1st percentile value.

This prevents extremes from distorting the mean while keeping our dataset size intact.



| The Pragmatic Approach



Iterative Testing

Data science is often about trial and error.

Don't blindly apply one rule!

Run your analysis or train your model three ways: keeping outliers, deleting outliers, and capping outliers. Compare the final results to see which yields the highest accuracy.